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Contents

CHAPTER ONE: INTRODUCTION 5
1.1 Background of the Study 5
1.2 Problem Statement 5
General Objective 6
Specific Objectives 6
1.4 Significance of the Study 6
Customers 6
Vendors 6
Internet Service Providers 6
Educational Value 6
CHAPTER TWO: SYSTEM REQUIREMENTS 7
2. System Requirements 7
2.1 Functional Requirements 7
2.1.1 User Authentication 7
2.1.2 Voucher Generation 7
2.1.3 Voucher Purchase 7
2.1.4 Voucher Storage 7
2.1.5 Sales Tracking 7
2.1.6 Voucher Management 7
2.1.7 Data Retrieval and Display 8
2.2 Non-Functional Requirements 8
2.2.1 Performance Requirement 8
2.2.2 Reliability Requirement 8
2.2.3 Security Requirement 8
2.2.4 Usability Requirement 8
2.2.5 Scalability Requirement 8
2.2.6 Availability Requirement 8
2.2.7 Maintainability Requirement 8
CHAPTER THREE: SYSTEM ARCHITECTURE AND DESIGN 9
3.1 Overall System Design 9
System Operation Flow 10
3.2 Client–Server Architecture 10

Client Side 10
Server Side 10
Architecture Communication Flow 11
3.3 Database Schema (Entity Relationship Diagram – ERD) 11
User Entity 11
Voucher Entity 11
Sales Entity 12
Relationship Between Entities 12
3.4 Authentication Mechanism 12
3.5 Security Measures Used 13
Access Control 13
Data Validation 13
Secure Data Storage 13
Server-Side Processing 13
Network Security 13
Future Security Improvements 13
Figure 3.1: System Architecture Diagram 14
Figure 3.2: Use Case Diagram 14
Figure 3.3: Database Schema Diagram 14
CHAPTER FOUR: IMPLEMENTATION DETAILS 16
4.1 Frontend Implementation 16
4.2 Backend Implementation 16
4.3 Database Implementation 16
4.4 API Integration 17
CHAPTER FIVE: SECURITY CONSIDERATIONS 18
5.1 User Authentication 18
5.2 Data Protection 18
5.3 Input Validation 18
5.4 Access Control 18
5.5 Future Security Improvements 18
CHAPTER SIX: TESTING AND RESULTS 19
6.1 Functional Testing 21
6.2 Integration Testing 21

6.3 User Testing 22
6.4 Overall Results 22
CHAPTER SEVEN: FUTURE WORK 23
REFERENCES 24

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Internet connectivity has become one of the most important services in modern society. It supports communication, education, business operations, entertainment, and access to information. Many Internet Service Providers (ISPs) use prepaid voucher-based systems to manage internet access, especially in developing countries where prepaid services are more affordable and easier to manage.

In voucher-based systems, customers purchase a voucher code which allows them to access the internet for a specific period or data amount. These vouchers are commonly generated and managed using network devices such as MikroTik routers. However, in many cases, voucher generation and sales are handled manually by vendors, which creates several challenges including slow service, human errors, poor record keeping, and difficulty in tracking daily or monthly sales.

With the advancement of web and mobile technologies, it is now possible to automate voucher generation, sales tracking, and customer interaction through digital systems. Automation improves efficiency, accuracy, and customer convenience.

The SmartConnect Voucher Management System was developed to provide an automated solution that allows customers to easily purchase internet vouchers through a web or mobile application, while vendors and administrators can monitor voucher sales and system performance in real time. The system integrates with MikroTik routers to automatically generate and manage voucher access.

1.2 Problem Statement

Many internet voucher systems currently rely on manual processes, which create several problems:

First, customers often experience delays when purchasing vouchers because they must physically visit vendors or wait for manual voucher generation. This reduces customer satisfaction.

Second, vendors lack automated tools to monitor and record voucher sales, making it difficult to track income, generate reports, or analyze business performance.

Third, manual systems are prone to human error, including incorrect voucher generation, lost records, and inaccurate sales reporting.

Fourth, customers have limited access to voucher services outside business hours.

Therefore, there is a need for an automated voucher management system that improves efficiency, accuracy, accessibility, and sales tracking.

1.3 Objectives of the System

General Objective

To design and implement a web-based and mobile-based voucher management system that automates internet voucher generation and provides real-time sales tracking.

Specific Objectives

The specific objectives of the system are:

- To develop a system that allows customers to purchase internet vouchers easily.
- To automate voucher generation using MikroTik router integration.
- To store voucher and sales information securely in a database.
- To provide vendors with real-time sales tracking and reporting.
- To improve efficiency and reduce manual work.

1.4 Significance of the Study

This system provides several benefits:

Customers

Customers can purchase vouchers quickly and conveniently without manual delays.

Vendors

Vendors can monitor voucher sales, track income, and manage business operations more efficiently.

Internet Service Providers

ISPs can improve service quality, customer satisfaction, and operational efficiency.

Educational Value

The system demonstrates practical application of web development, database management, and network integration technologies.

CHAPTER TWO: SYSTEM REQUIREMENTS

2. System Requirements

System requirements describe the conditions and capabilities needed for the SmartConnect Voucher Management System to operate effectively. These requirements are divided into **functional requirements** and **non-functional requirements**.

2.1 Functional Requirements

Functional requirements describe the specific services and operations that the system must perform. These define how the system behaves and what tasks it must accomplish.

2.1.1 User Authentication

The system shall allow users (vendors or administrators) to log in using a username and password. This ensures that only authorized users can access system features and manage vouchers or sales information.

2.1.2 Voucher Generation

The system shall automatically generate internet voucher codes by communicating with the MikroTik router through an API. This eliminates manual voucher creation and reduces human error.

2.1.3 Voucher Purchase

The system shall allow customers to select and purchase available internet vouchers through the web or mobile interface. After purchase, the voucher code shall be displayed to the user.

2.1.4 Voucher Storage

The system shall store voucher details such as voucher code, duration, price, and creation date in the database. This ensures proper record keeping and easy retrieval.

2.1.5 Sales Tracking

The system shall record all voucher sales and store transaction information such as date, voucher type, and amount. Vendors shall be able to view daily and monthly sales reports.

2.1.6 Voucher Management

The system shall allow administrators or vendors to manage vouchers, including viewing, generating, and monitoring voucher usage.

2.1.7 Data Retrieval and Display

The system shall allow users to view voucher information, sales reports, and system records through the interface.

2.2 Non-Functional Requirements

Non-functional requirements describe the system's quality attributes and performance characteristics.

2.2.1 Performance Requirement

The system shall respond to user requests within a short time (less than 3 seconds). This ensures efficient operation and good user experience.

2.2.2 Reliability Requirement

The system shall operate continuously without frequent failures. The database shall store information accurately and prevent data loss.

2.2.3 Security Requirement

The system shall protect user data from unauthorized access by requiring login authentication and secure database storage.

2.2.4 Usability Requirement

The system shall provide a simple and user-friendly interface so that users can easily purchase vouchers and view information without technical knowledge.

2.2.5 Scalability Requirement

The system shall support an increasing number of users and vouchers without reducing performance. This allows system expansion as business grows.

2.2.6 Availability Requirement

The system shall be accessible at all times when internet connection is available. This ensures customers can purchase vouchers anytime.

2.2.7 Maintainability Requirement

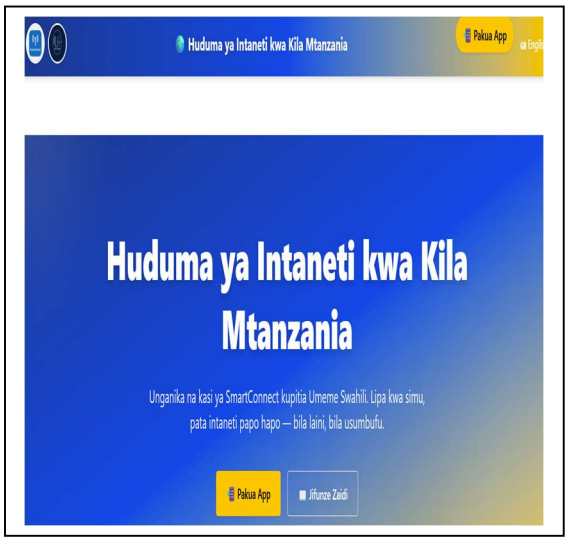
The system shall be easy to update, modify, and maintain by developers. This ensures future improvements can be implemented easily.

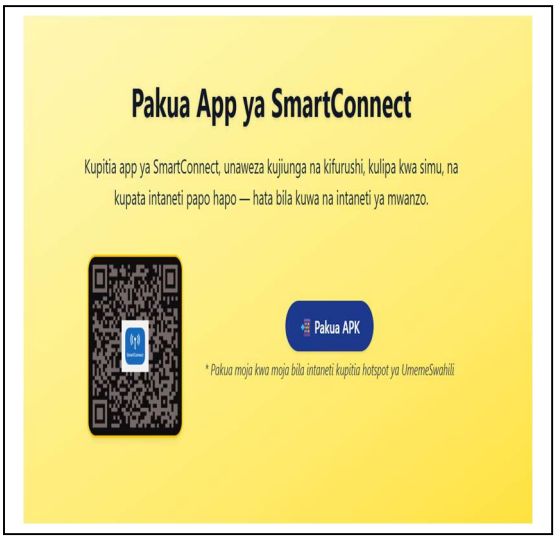
CHAPTER THREE: SYSTEM ARCHITECTURE AND DESIGN

System architecture describes the structure of the system, its components, and how those components interact with each other. The SmartConnect Voucher Management System uses a client–server architecture integrated with a database and MikroTik router to provide automated voucher management services.

3.1 Overall System Design

The SmartConnect Voucher Management System is designed as a web-based and mobile-accessible application that allows customers to purchase internet vouchers and allows vendors or administrators to monitor and manage voucher sales.





The system consists of four main components:

Client Interface
This is the user interface accessed through a web browser or mobile application. It allows users to interact with the system, purchase vouchers, and view information.

Application Server
The server is responsible for processing user requests, managing system logic, generating vouchers, and communicating with the database and MikroTik router. The server is developed using Node.js and Express.js.

Database System
The database stores system data such as user accounts, voucher codes, and sales transactions. MongoDB or Firebase Firestore is used to store and manage this data securely.

Network Router (MikroTik)
The MikroTik router generates voucher codes and controls internet access based on voucher authentication.

System Operation Flow

The system operates in the following sequence:

1. The user accesses the system using a web browser or mobile application.
2. The user selects a voucher package.
3. The client sends a request to the application server.
4. The server processes the request.
5. The server communicates with the database to store or retrieve information.
6. The server communicates with the MikroTik router to generate a voucher.
7. The voucher is sent back to the user.
8. The user enters the voucher to access the internet.

This design ensures efficient and automated system operation.

3.2 Client–Server Architecture

The SmartConnect system uses a client–server architecture. This architecture separates the user interface from the system processing and database storage.

Client Side

The client side includes:

- Web browser
- Mobile application

Functions performed by the client include:

- Sending requests to the server
- Displaying voucher information
- Displaying system responses

Technologies used include:

- HTML
- CSS
- JavaScript
- Flutter (mobile)

Server Side

The server performs the main system processing.

Functions include:

- Processing user requests
- Generating vouchers
- Managing database communication

- Managing authentication

Technologies used include:

- Flutter
- Firebase

Architecture Communication Flow

Client → Server → Database
Client → Server → MikroTik Router
Server → Client (Response)

This architecture improves system performance, scalability, and security.

3.3 Database Schema (Entity Relationship Diagram – ERD)

The database is designed to store all system information in structured form.

The main entities include:

User Entity

This entity stores system user information.

Attributes include:

- user_id (Primary Key)
- username
- password
- role

Purpose:
To identify and authenticate users.

Voucher Entity

This entity stores voucher information.

Attributes include:

- voucher_id (Primary Key)
- voucher_code
- price
- duration
- status

Purpose:
To store and manage voucher details.

Sales Entity

This entity stores transaction information.

Attributes include:

- sale_id (Primary Key)
- voucher_id (Foreign Key)
- date
- amount

Purpose:
To track voucher sales.

Relationship Between Entities

One voucher can appear in one sales record.

The relationship allows the system to track which voucher was sold and when.

This database structure ensures efficient data storage and retrieval.

3.4 Authentication Mechanism

Authentication is used to verify user identity before granting access to the system.

The system uses username and password authentication.

The authentication process works as follows:

1. User enters username and password.
2. System sends credentials to server.
3. Server checks credentials in database.
4. If valid, access is granted.
5. If invalid, access is denied.

This ensures that only authorized users can access system features.

Future improvements may include:

- Password encryption
- Token-based authentication
- Multi-factor authentication

3.5 Security Measures Used

Security is important to protect system data and prevent unauthorized access.

The following security measures are implemented:

Access Control

Only authorized users can access system features.

This prevents unauthorized system use.

Data Validation

The system validates all user input.

This prevents invalid or harmful data.

Secure Data Storage

Data is stored securely in the database.

This prevents data loss and unauthorized access.

Server-Side Processing

System logic is processed on the server.

This prevents users from manipulating system operations.

Network Security

Communication between system components is controlled and monitored.

This prevents unauthorized network access.

Future Security Improvements

Future security enhancements include:

- Password hashing
- HTTPS encryption
- Firewall protection
- Secure login sessions

Figure 3.1: System Architecture Diagram

Place the **System Architecture** image under the **System Architecture section** and write:

Figure 3.1: System Architecture of the SmartConnect Voucher Management System

This figure shows how the user interacts with the system through a mobile app or web browser. The request is sent to the application server built using Node.js and Express.js, which processes the request and communicates with the database and MikroTik router to generate and manage vouchers.

Figure 3.2: Use Case Diagram

Place the **Use Case Diagram** under the **System Design or System Architecture section** and write:

Figure 3.2: Use Case Diagram of the SmartConnect System

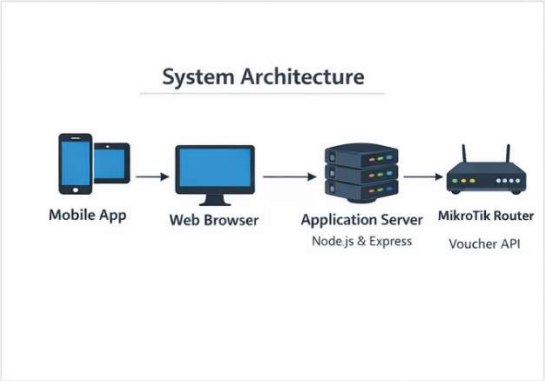
This diagram illustrates how customers, vendors, and administrators interact with the system. Customers purchase vouchers, vendors monitor sales, and administrators manage users and system operations.

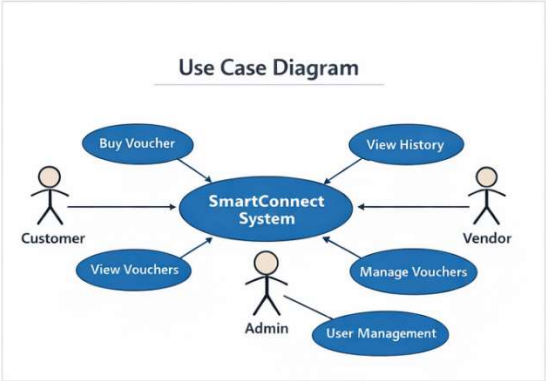
Figure 3.3: Database Schema Diagram

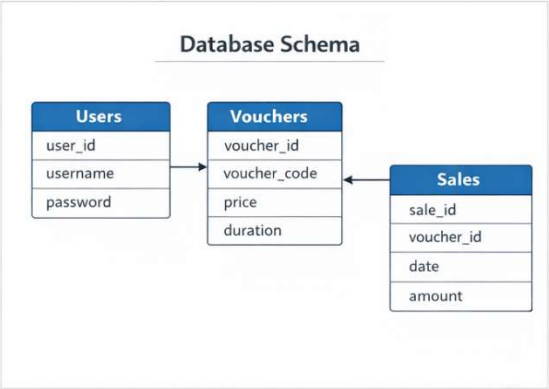
Place the **Database Schema** image under the **Implementation Details or Database Design section** and write:

Figure 3.3: Database Schema of the SmartConnect System

This diagram shows how data is stored in the system database, including users, vouchers, and sales tables, and how they are related.


System Architecture
Mobile App
Web Browser
Application Server
Node.js & Express
MikroTik Router
Voucher API


Use Case Diagram
Buy Voucher
View History
Customer
SmartConnect System
Vendor
View Vouchers
Manage Vouchers
Admin
User Management


Database Schema
Users
user_id
username
password
Vouchers
voucher_id
voucher_code
price
duration
Sales
sale_id
voucher_id
date
amount

CHAPTER FOUR: IMPLEMENTATION DETAILS

Implementation describes how the system was built.

4.1 Frontend Implementation

The frontend provides user interaction.

Technologies used include:

- Flutter Technology
- Flutter for styling
- Flutter for functionality
- Flutter for mobile interface

Frontend allows users to:

- View vouchers
- Purchase vouchers
- View information

4.2 Backend Implementation

Backend handles system operations.

It was developed using:

- Flutter
- Firebase

Backend functions include:

- Processing user requests
- Communicating with database
- Generating vouchers
- Sending responses

4.3 Database Implementation

Database stores system information.

Main data includes:

Users table:

- User ID
- Username
- Password

Voucher table:

- Voucher code
- Duration
- Price

Sales table:

- Sale ID
- Date
- Amount

4.4 API Integration

The system connects to MikroTik router.

This allows:

- Automatic voucher generation
- Automated access control

This eliminates manual work.

CHAPTER FIVE: SECURITY CONSIDERATIONS

Security protects system and user data.

5.1 User Authentication

Users must login before accessing system features.

This prevents unauthorized access.

5.2 Data Protection

Data is stored securely in database.

Prevents data loss.

5.3 Input Validation

System checks user input.

Prevents errors and attacks.

5.4 Access Control

Only authorized users can perform certain actions.

Improves security.

5.5 Future Security Improvements

Future improvements include:

- Password encryption
- Secure communication (HTTPS)
- Multi-factor authentication

CHAPTER SIX: TESTING AND RESULTS

Testing ensures system works correctly.

Figure 1: Login page, this is where user creates and log into his or her account.

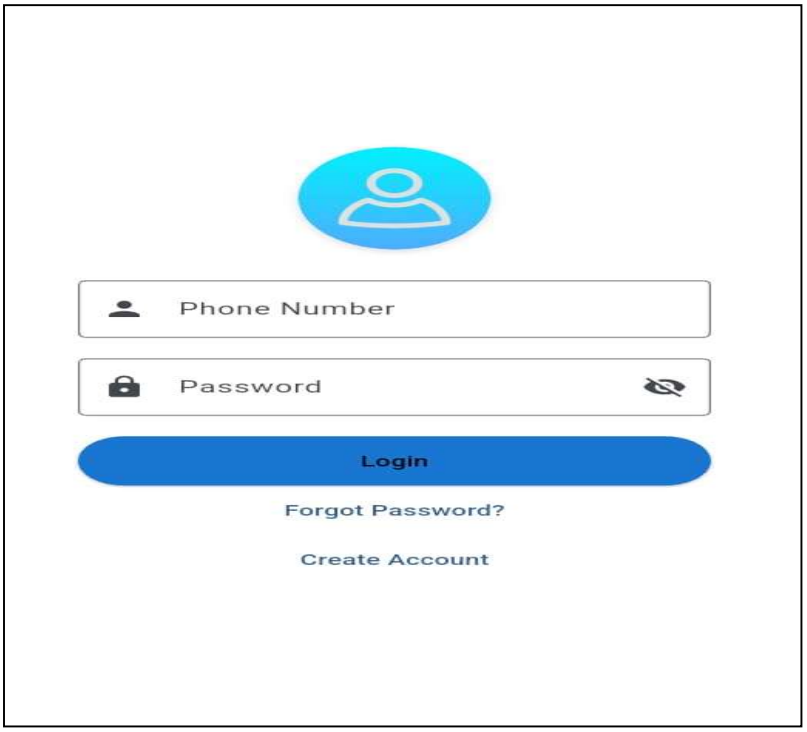
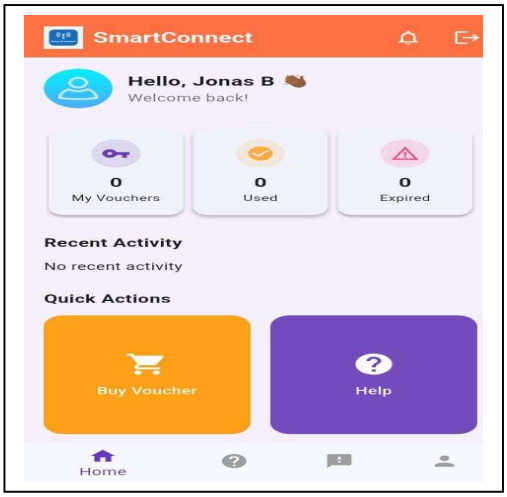


Figure 2: SmartConnect front page after the user logs into and profile page.



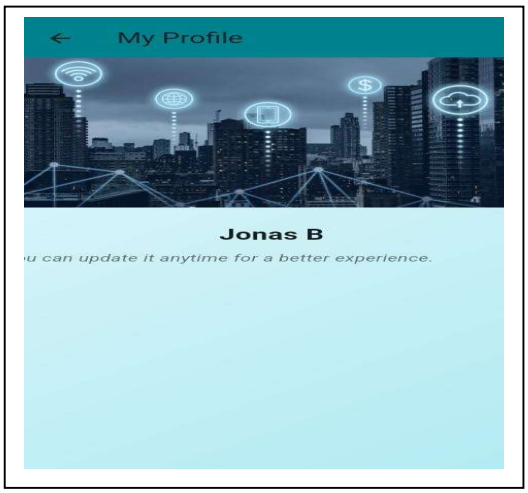


Figure 3: User buys voucher.

Figure 4: Payment is generated and choosing payment methods for mobile either MPESA, AIRTELMONEY, etc.

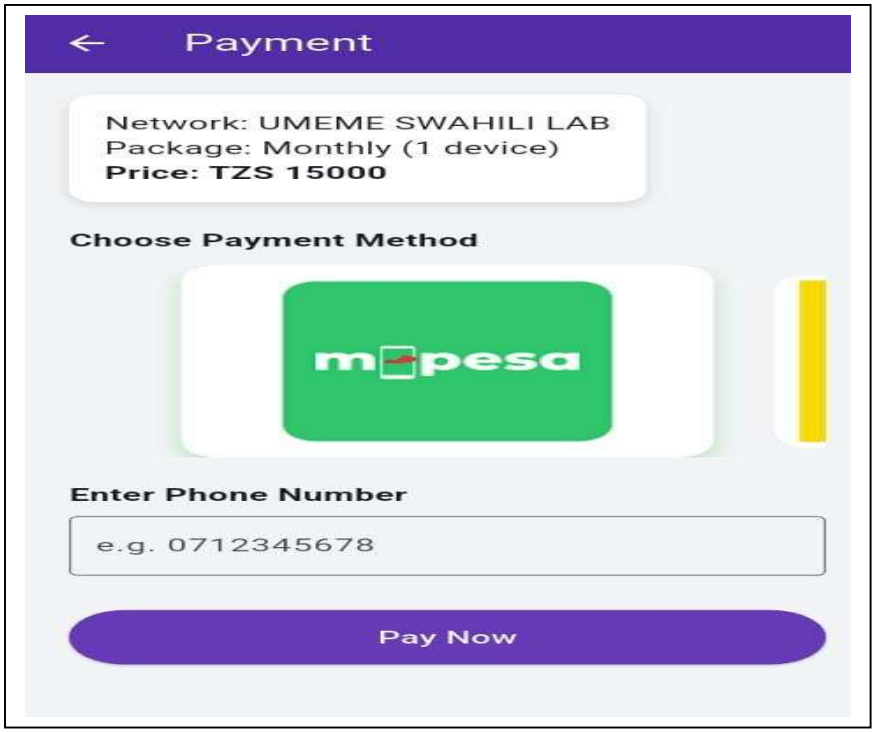
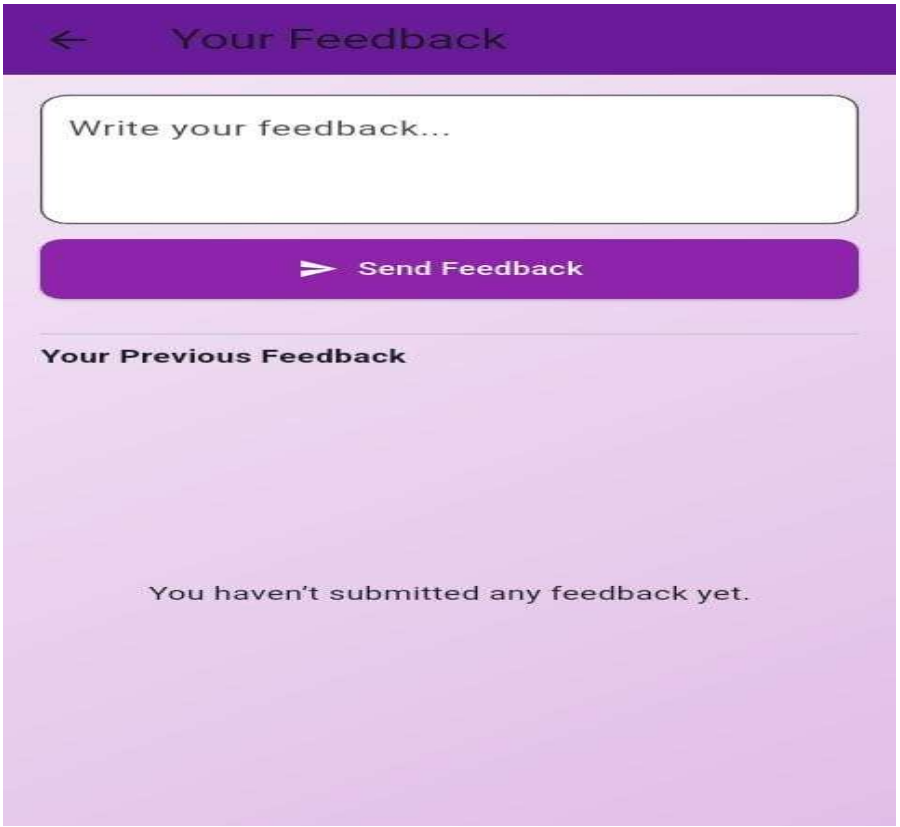


Figure 5: User can write the feedback.



6.1 Functional Testing

Tested system features:

Voucher generation
Result: Successful

Voucher storage
Result: Successful

Report display
Result: Successful

6.2 Integration Testing

Tested communication between:

Application and database
Result: Successful

Application and MikroTik router
Result: Successful

6.3 User Testing

Users tested system.

Results showed:

System is easy to use
System is fast
System improves efficiency

6.4 Overall Results

System achieved objectives.

It successfully automated voucher management.

CHAPTER SEVEN: FUTURE WORK

Future improvements include:

Mobile money integration such as M-Pesa, Airtel Money, and Tigo Pesa.

Cloud deployment to allow access from anywhere.

Improved reporting tools.

Improved security features.

Mobile application expansion.

User account management features.

REFERENCES

Flutter Documentation
MikroTik RouterOS Documentation
Firebase Documentation
Web development textbooks and tutorials